Using Existing LLMs for Explainable Emotion Analysis/Classification

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Abstract

Large Language Models (LLMs) have demonstrated strong performance in natural language understanding tasks, including emotion recognition, but they're a "black box," so we don't really know how they make their decisions. This can reduce trust and usability in affective computing applications. This research attempts to address this challenge by employing the locally executed open-source LLMs for interpretable emotion analysis. The key idea is to exploit the generative power of the models not only for emotion classification of text but also at the same time generate human-understandable justifications for their predictions. At the end the performance of the models will be assessed both quantitatively and qualitatively. During the study we ran LLMs locally, designed prompts to classify emotion from text and explain its reasoning. Models process text from a labeled dataset to generate a corpus of emotion labels with corresponding explanations. The performance is assessed with accuracy, F1-scores, Cohen's kappa, MCC and confusion matrices.

1 Introduction

This section introduces the core concepts underpinning our research. We begin by outlining the task of emotion classification and its inherent complexities. We then discuss the recent paradigm shift towards using Large Language Models (LLMs) for this task, followed by a brief overview of relevant evaluation metrics. Finally, we establish the critical need for Explainable AI (XAI) and introduce Chain-of-Thought prompting as our chosen method to achieve it.

1.1 Emotion Classification

Emotion is central to human intelligence and should have a similar role in Artificial Intelligence (AI) [3]. The automated classification of emotion from data is a key task in the field of Affective Computing. However, the task is notoriously difficult due to the complex nature of emotion itself, which has deep evolutionary roots and serves a wide range of adaptive functions [9].

Emotion is conveyed through multiple modalities, including facial expressions, speech, and body language. While analysis often centers on facial cues [6], research has shown that in moments of intense feeling, body language can be more discriminative than facial expressions [1]. Furthermore,

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subtle and spontaneous 'micro-gestures' have been identified as reliable indicators of hidden emotions, acting as "honest signals" of a person's true internal state [4, 8].

In the text domain, a major challenge is the sheer breadth of human emotional experience. Traditional approaches often classify text into a small set of basic emotions, but this fails to capture the nuance of expression. Psychological studies suggest humans can experience up to 34,000 distinct emotional states [7, 9]. Addressing this complexity requires large-scale datasets with fine-grained labels, such as the GoEmotions dataset, which contains 58,000 comments annotated for 27 emotion categories [5].

1.2 LLMs for Emotion Classification

The field of Natural Language Processing (NLP) has recently shifted towards the use of pre-trained Large Language Models (LLMs) [2]. Previously, achieving strong performance on a task required fine-tuning a model on a large, task-specific dataset, often containing thousands of labeled examples [2]. This approach is costly and inflexible. In contrast, humans can typically learn a new language task from a few simple instructions or a handful of examples—a capability that AI systems have struggled to replicate [2].

The seminal work on GPT-3 demonstrated that scaling up language models greatly improves their ability to perform tasks in a **few-shot** setting [2]. This method, known as "in-context learning," allows a model to perform a new task by conditioning it on a prompt containing a few demonstrations, with no gradient updates or fine-tuning required. This includes **zero-shot** (instruction only) and **one-shot** (one example) settings [2]. This flexibility is perfectly suited to the challenge of emotion classification, as it enables an "open-vocabulary" approach where the model is not confined to a predefined set of labels and can better capture the nuanced spectrum of human feelings [7].

1.3 Evaluation Metrics

To quantitatively assess the performance of our models on the emotion classification task, we will employ standard metrics from the machine learning literature. These include **Accuracy**, which measures the overall proportion of correct predictions, as well as **Precision**, **Recall**, and the **F1-Score**, which provide a more nuanced view of performance, especially in cases of class imbalance [5]. For our multi-label task, we will focus on the weighted-average F1-Score. Additionally **Balanced Accuracy** is used to evaluate the accuracy adjusted for class distribution, **Matthews Correlation Coefficient (MCC)** for a robust measure considering all confusion matrix elements, **Cohen's Kappa** to measure the agreement accounting for chance and **Per-emotion F1-scores** to get class-specific performance for each of six emotions.

While these metrics effectively evaluate classification accuracy, assessing the quality of the generated explanations is more challenging and will be handled through qualitative analysis, which is discussed in the Experiments section.

1.4 Explainable AI

A significant limitation of many powerful AI models is their "black box" nature; while they can make highly accurate predictions, their internal decision-making processes are opaque [3, 10]. This lack of transparency impedes trust, makes it difficult to debug errors, and limits the utility of these models in high-stakes or human-centered applications. In response, there has been a growing call for the development of "explanatory models" that provide not just predictions, but also interpretable and actionable insights into their reasoning [10].

Our project adopts this goal by focusing on an explainability technique known as **Chain-of-Thought (CoT) prompting**. CoT is a simple but effective method where a few examples of step-by-step reasoning are included in a model's prompt [12]. This encourages the LLM to generate its own intermediate reasoning steps before arriving at a final answer for a new problem. This generated text provides an interpretable window into the model's behavior, serving as a direct explanation for its classification and transforming it from an opaque black box into a more transparent system [12].

2 Related Works

Our work builds on ideas from three main areas: how we understand emotion itself, how we can use new AI like Large Language Models (LLMs) for language tasks, and how we can make that AI explain its own reasoning.

The study of emotion recognition is built on a complex foundation. Emotions themselves have deep evolutionary roots that help organisms survive, which is part of why they are so varied and nuanced [9]. While we often talk about a handful of "basic" emotions, some researchers estimate that humans can express as many as 34,000 distinct emotional states [7, 9]. This huge range makes classifying them a real challenge. Traditionally, researchers have looked for emotional cues in facial expressions, following the foundational work of Ekman [6]. But this isn't the whole picture. During intense moments, for example, body language can actually be a better clue than facial expressions for telling positive and negative emotions apart [1]. Even more subtle are 'micro-gestures'—small, involuntary movements that can act as "honest signals" of what someone is truly feeling, even when they're trying to hide it [4, 8]. When we turn to text, the problem of limited categories becomes even clearer. To get past this, the idea of **Open-Vocabulary Multimodal Emotion Recognition (OV-MER)** has been proposed, which pushes for models that are not stuck with a short, predefined list of emotions [7]. Building such models requires rich data, which is where large-scale datasets like **GoEmotions** come in, offering 58,000 text comments labeled with 27 different emotion categories [5].

This need for more flexible, open-vocabulary models is perfectly met by a recent shift in Natural Language Processing (NLP). Instead of training a new model for every single task, the field has moved towards using massive, pre-trained Large Language Models. As the groundbreaking work on GPT-3 showed, these models can perform new tasks with little to no training data [2]. They can work in a **zero-shot** setting with just an instruction, or in a **few-shot** setting with a handful of examples given in the prompt [2]. This ability to learn 'in-context' is what makes the open-vocabulary approach to emotion analysis possible.

But there's a major catch with these powerful LLMs: they are often "black boxes" [3, 10]. We can see the input and the output, but the reasoning process in between is hidden. This is a huge problem for trust, especially when dealing with something as subjective as emotion. This has led to a push for 'explanatory learner models'—systems that are designed not just to be accurate, but to provide clear and useful insights into their own logic [10]. To tackle this, our project focuses on a technique called **Chain-of-Thought (CoT) prompting**. The idea is simple but powerful: by showing the model a few examples of step-by-step reasoning, it learns to generate its own reasoning process before giving a final answer [12]. That generated reasoning becomes the explanation we're looking for, giving us a window into the model's 'thinking' and making it more transparent.

3 Implementation Details

3.1 Dataset

For this research, we are using the dair.ai/emotion dataset. The dataset was introduced in the 2018 published paper "CARER: Contextualized Affect Representations for Emotion Recognition" by Saravia et al. [11] The dataset is available in two different versions, split and unsplit. The split version, which we are using for this research, contains 20,000 English Twitter messages labeled in six different basic emotion categories (anger, fear, joy, love, sadness, and surprise). In the original paper, the dataset included eight different emotion categories. Messages in the dataset are pre-split into training (16,000 messages), validation (2,000 messages), and testing (2,000 messages) sets. The unsplit version includes 416,809 labeled messages without predefined split. Distribution of different emotion classes is introduced in Table 1.

4 Few-shot Learning Experimental Design

We investigated the performance of the following four Large Language Models Mistral 7B, Qwen2.5 7B, FLAN-T5 small and FLAN-T5 base in the emotion classification problem under varying few-shot configurations and tweet length thresholds. The selection of these models made it possible to compare how similar-sized generative models from different architectures (Mistral vs Qwen), and models of

Table 1: Distribution of emotion categories in the dataset.

Label	Emotion	Count	Percentage
1	Joy	5,362	33,5%
0	Sadness	4,666	29,2%
3	Anger	2,159	13,5%
4	Fear	1,937	12,1%
2	Love	1,304	8,2%
5	Surprise	572	3,6%

different parameter scales within the same architecture family (T5-small vs T5-base) perform on this classification task. This comparative framework allowed for evaluation of both model size and architectural differences. The dataset described in section 3.1 was used in the evaluation, and all models were run locally. To ensure robust evaluation while maintaining computational feasibility, stratified random sampling was employed to select 500 test samples, preserving the original class distribution to prevent biased evaluation. This sampling strategy ensured that the emotion classes were originally distributed while allowing a comprehensive model comparison.

4.1 Experimental Design

The evaluation implemented three distinct learning paradigms, which were zero-shot, 3-shot and 5-shot learning. Details about these methods have been explained in section 1.2

Few-shot examples were pre-selected using a balanced sampling strategy that ensured coverage of all emotion classes while respecting tweet length constraints. The examples were formatted using a consistent prompt template that instructed models to classify the predominant emotion, with deterministic generation settings (temperature=0.0, seed=42) to ensure reproducibility.

Another aspect of this evaluation was the systematic investigation of three tweet length thresholds (100, 200, and 500 characters) for few-shot example selection. Shorter examples reduced prompt length, enabling more efficient inference. We also wanted to see if example verbosity affected the model learning in few-shot scenarios.

4.2 Implementation Details

Models (Mistral 7B, Qwen2.5 7B) were accessed through the Ollama API with deterministic generation parameters: temperature=0.0, top p=1.0, top k=1, repeat penalty=1.0, and a maximum of 10 predicted tokens. Parallel processing with 4 concurrent workers (ThreadPoolExecutor) was implemented to accelerate inference across the 500 test samples.

HuggingFace Models (FLAN-T5 small, FLAN-T5 base) were downloaded locally and loaded using the HuggingFace Transformers pipeline with deterministic settings: do sample=False, num. beams=1 and max length=10. GPU acceleration was utilized when available (CUDA), with batch processing (batch size=8 for 3-shot, batch size=16 for others) to optimize throughput.

Outputs of the models were post-processed to extract the predicted emotion label by matching against the six target emotions (Table 1). This approach manages how these models generate natural language while keeping the evaluation process consistent and fair.

4.3 Results

Summary of performance details of all models used in few-shot experiments are presented in table 2.

Table 2: Summary of Performance Evaluation

Model	Best Config	Accuracy	F1-Weighted	F1-Macro	Key Insight
FLAN-T5 Base (80m)	3-shot, 100 chars	70.6%	0.697	0.550	Best overall
Qwen2.5 7B	5-shot, 100 chars	59.8%	0.604	0.148	Best Ollama
Mistral 7B	3/5-shot, 100 chars	53.0%	0.567	0.046	Poor macro F1
FLAN-T5 Small (250m)	Zero-shot	41.0%	0.419	0.370	Few-shot hurts

Model-specific results are listed below.

Mistral 7B:

- Zero-shot: 49.6% accuracy (F1-weighted: 0.5247, F1-macro: 0.0328) - consistent across all thresholds (100,200,500).
- 3-shot @ 100 chars: 53.0% accuracy (F1-weighted: 0.5592, F1-macro: 0.0453)
- 5-shot @ 100 chars: 53.0% accuracy (F1-weighted: 0.5666, F1-macro: 0.0459)
- Increasing threshold lenght reduced performance (100→500 chars: ACC 53.0% → 48.0% for 5-shot) (Figure 2)

Qwen2.5 7B:

- Zero-shot: 54.8% accuracy (F1-weighted: 0.5501, F1-macro: 0.0741) - consistent across all thresholds
- 3-shot @ 100 chars: 58.8% accuracy (F1-weighted: 0.5948, F1-macro: 0.1237)
- 5-shot @ 100 chars: 59.8% accuracy (F1-weighted: 0.6040, F1-macro: 0.1476) [BEST for Ollama]
- More stable performance across threshold variations (Figure 3)

FLAN-T5 Small:

- Zero-shot: 41.0% accuracy (F1-weighted: 0.4191, F1-macro: 0.3698) [BEST for this model]
- 3-shot @ 100 chars: 32.4% accuracy (F1-weighted: 0.3369, F1-macro: 0.2500)
- 5-shot @ 100 chars: 34.0% accuracy (F1-weighted: 0.3444, F1-macro: 0.2655)
- Lower performance with few-shot examples

FLAN-T5 Base:

- Zero-shot: 68.0% accuracy (F1-weighted: 0.6739, F1-macro: 0.5171)
- 3-shot @ 100 chars: 70.6% accuracy (F1-weighted: 0.6971, F1-macro: 0.5500) [BEST overall]
- 5-shot @ 200 chars: 70.4% accuracy (F1-weighted: 0.6977, F1-macro: 0.5592)
- Best overall performance across different configurations

Statistical Performance Across Configurations Ollama Models Overall Statistics:

- Mean accuracy across all configurations: 53.5% (SD: 4.17%)
- Mean F1-weighted: 0.5593 (SD: 0.0224)
- Mean F1-macro: 0.0890 (SD: 0.0531)
- Performance range: 46.6% to 59.8%

HuggingFace Models Overall Statistics:

- Mean accuracy across all configurations: 53.3% (SD: 16.73%)
- Mean F1-weighted: 0.5281 (SD: 0.1643)
- Mean F1-macro: 0.4376 (SD: 0.1494)
- Performance range: 32.4% to 70.6%

Key Observations

We noticed that the impact of few-shot examples varied dramatically across model architectures. While Qwen2.5 7B showed improvement from zero-shot (54.8%) to 5-shot (59.8%), the FLAN-T5 Small model behaved exactly opposite with degraded performance in few-shot settings. This may suggest that smaller models lack the capacity to effectively leverage additional context, potentially

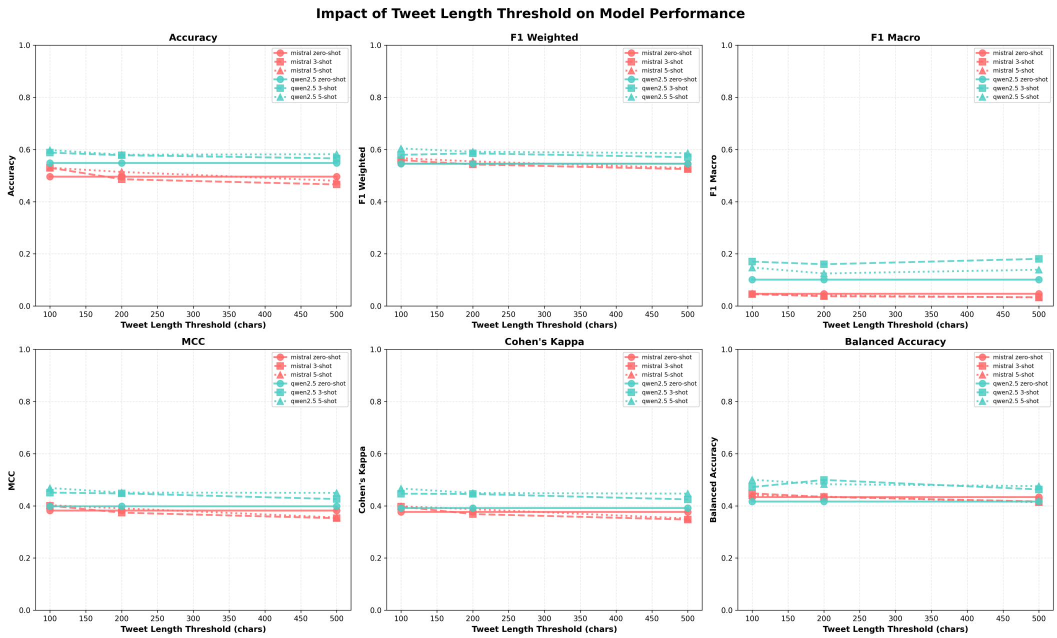

Figure 1: Impact of Tweet Lenght Threshold on Model Performance on Ollama Models.

suffering from what has been termed "early ascent" where limited examples retrieve incorrect patterns before sufficient task learning occurs.

Within the FLAN-T5 family, the base model (250M parameters, best accuracy (70.6%)) outperformed the small variant (80M parameters, best accuracy (41.0%)). This significant gap underscores the importance of model capacity in complex emotion recognition tasks. However, the Ollama models (both 7B parameters) showed more modest differentiation (Mistral accuracy (53%), Qwen2.5 accuracy (59.8%)). Suggesting that beyond a certain parameter threshold, architectural and training differences dominate performance variations.

There was also an impact with the tweet length. The 100-character threshold consistently yielded optimal or near-optimal performance across all models. For Mistral 7B, performance degraded (Accuracy 53.0% $\rightarrow$ 48.0%) when moving from 100 to 500-character examples in the 5-shot condition. (Figure 1). However, there was a slight increase in the F1-macro on the Qwen2.5 7B model, when using a higher threshold. Overall, the shorter, more informative-packed examples were more effective for emotion classification, potentially because shorter examples are typically more simple and enable better focus on emotion-relevant features.

FLAN-T5 Base (70.6% accuracy) substantially outperformed both Ollama models (Qwen2.5: 59.8%, Mistral: 53.0%), despite having significantly fewer parameters (250M vs 7B). This highlights that architectural design and task-specific training regimes can outweigh raw parameter count. The FLAN-T5 models' encoder-decoder architecture and text-to-text training paradigm appear particularly well-suited for emotion classification tasks, while the autoregressive, generative nature of Mistral and Qwen may introduce additional challenges in constrained classification scenarios.

The higher variance in HuggingFace models reflects the performance differential between the small and base versions, while the Ollama models exhibited more uniform behavior across configurations. In all models, the emotion "love" was the most difficult emotion to correctly predict (Figure 5) (Figure 11).

5 Chain-of-Thought reasoning

In the **Chain-of-Thought** or **CoT** portion of our project, we compared how using CoT reasoning would impact model performance and help improve the explainability of the classifications. For this part, we used two different models, Gemma3:270m and Gemma3:4b using the help of the Ollama

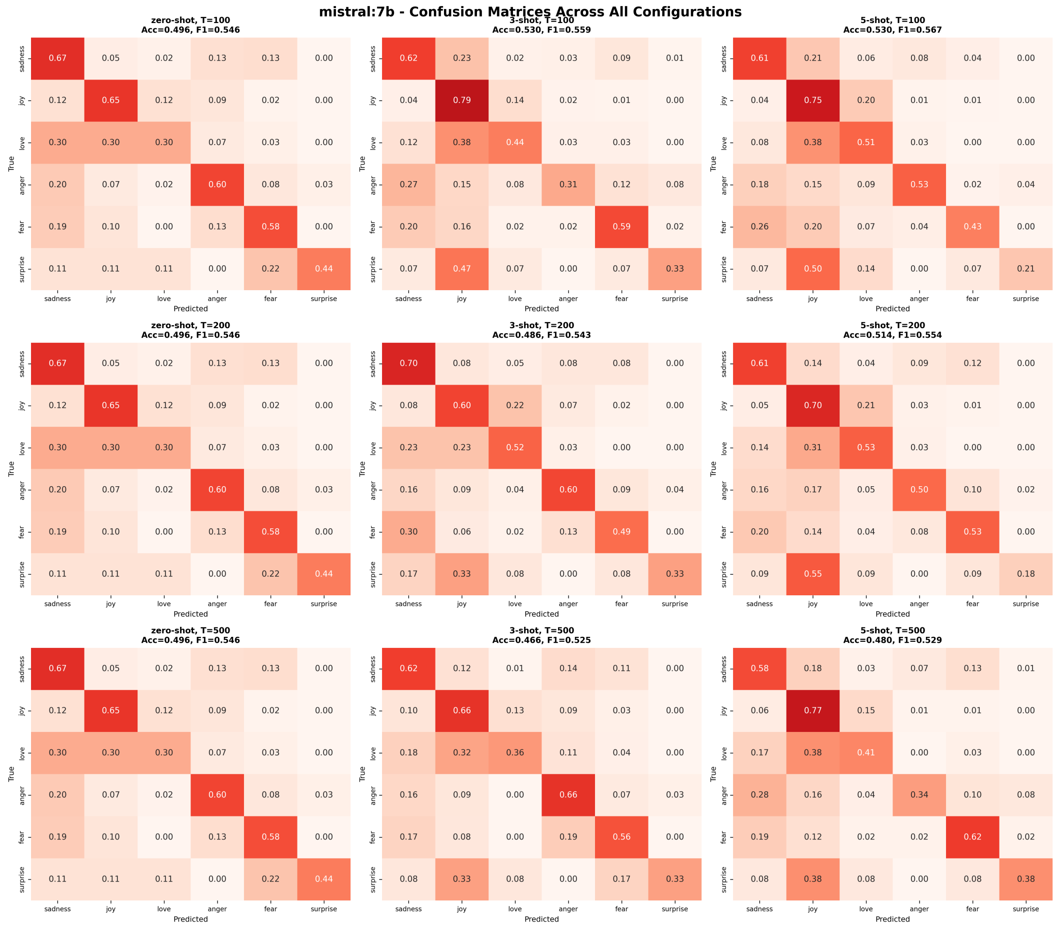

Figure 2: Mistral 7B confusion matrices for all 9 different configurations

python library. The model testing was carried out using 100 messages selected from the test set of the data set.

5.1 Experiment design

The LLM was asked to classify the given text into one of the possible categories using the following instructional promt: **"You are an expert sentiment analysis AI. Your task is to accurately analyze what emotion from the six base emotions (sadness, joy, love, anger, fear, surprise) a given text conveys."** To compare the results, the LLM was first asked to answer with one word only, the name of the emotion and then a second time with the additional instruction: **"Use chain-of-thought reasoning to determine which emotion of the six base emotions (sadness, joy, love, anger, fear, surprise) the text evokes."**

5.2 Examples

As an example, the model was tasked to label the following text with the correct emotion. **"I realized my mistake and I'm really feeling terrible and thinking that i shouldn't do that"** When the model is given the text, it should find reasoning and explain how it got to the conclusion of the predicted label. **"The phrases 'really feeling terrible' and 'i shouldn't do that' strongly indicate regret and remorse. The speaker is acknowledging a negative action and experiencing a negative emotional response to it. This aligns most closely with sadness and potentially fear (of repeating the mistake)."** In this example the predicted label came out as sadness, which was also the correct one.

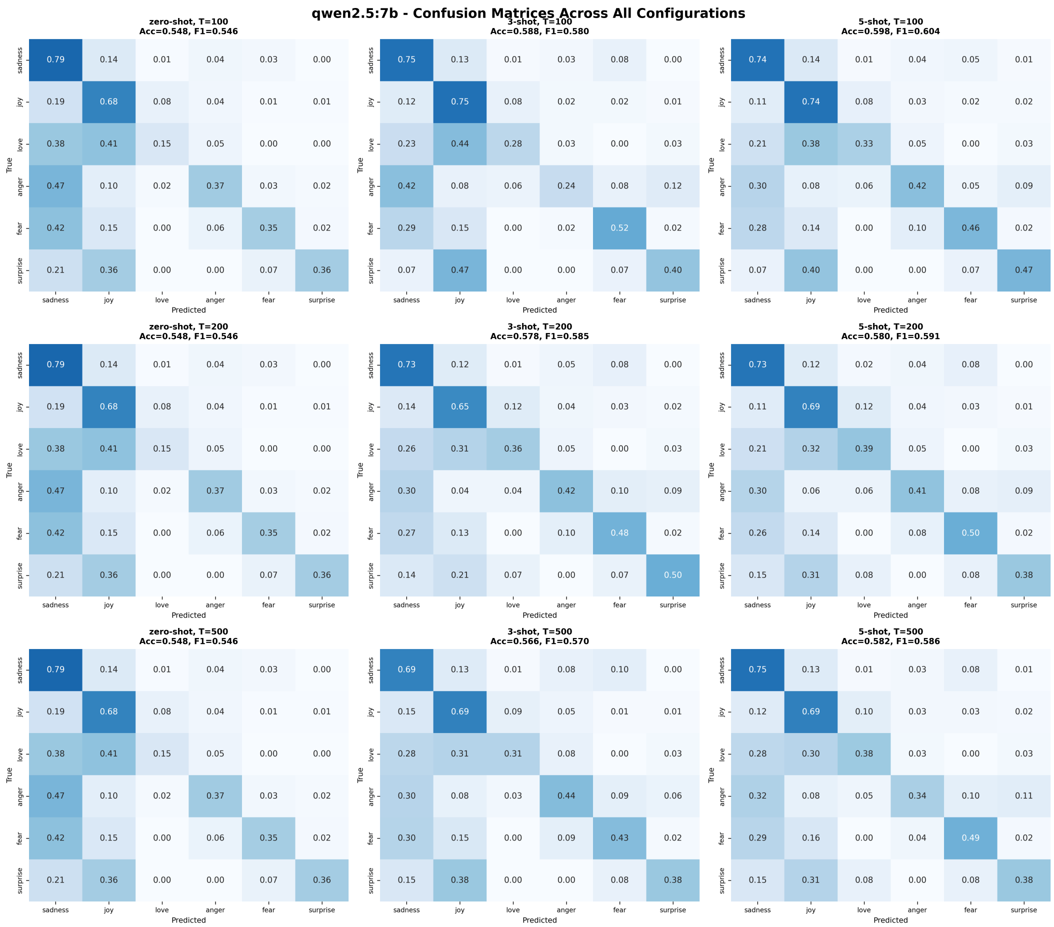

Figure 3: Qwen2.5 7B confusion matrices for all 9 different configurations

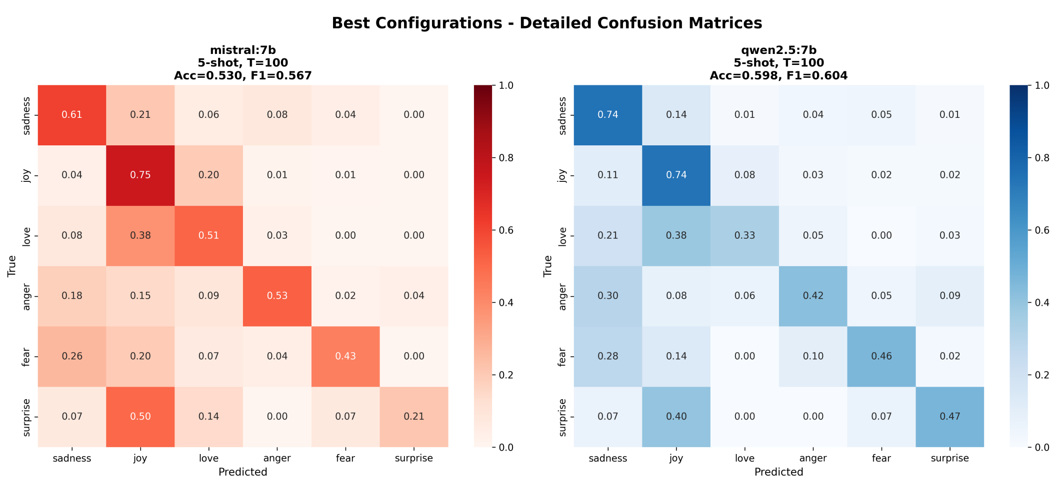

Figure 4: Side-by-side comparison of best configurations for Ollama models

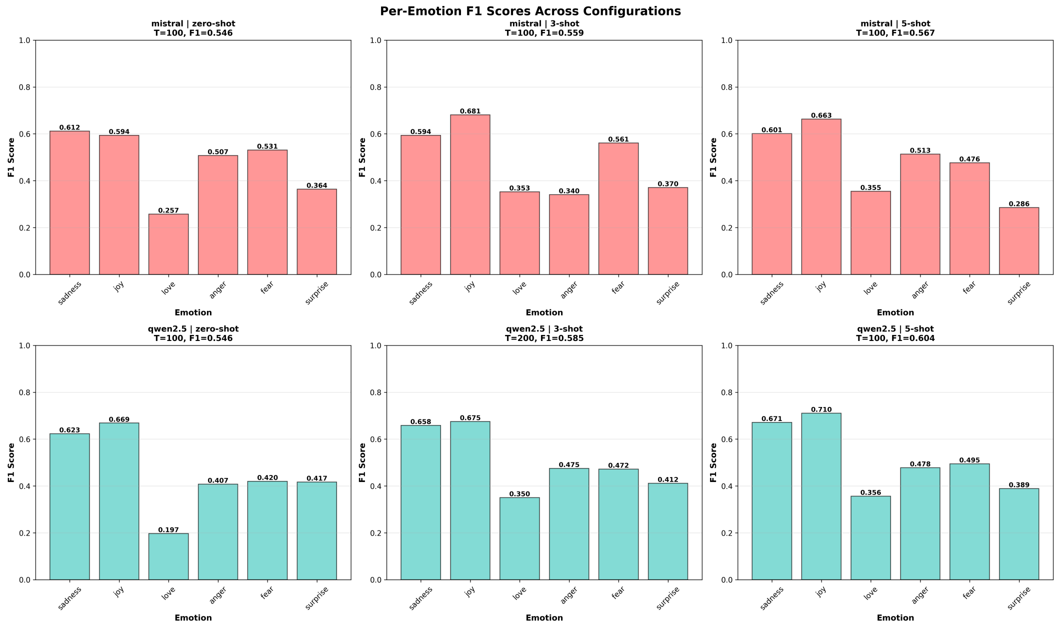

Figure 5: Per-emotion F1-score breakdown across configurations for Ollama Models

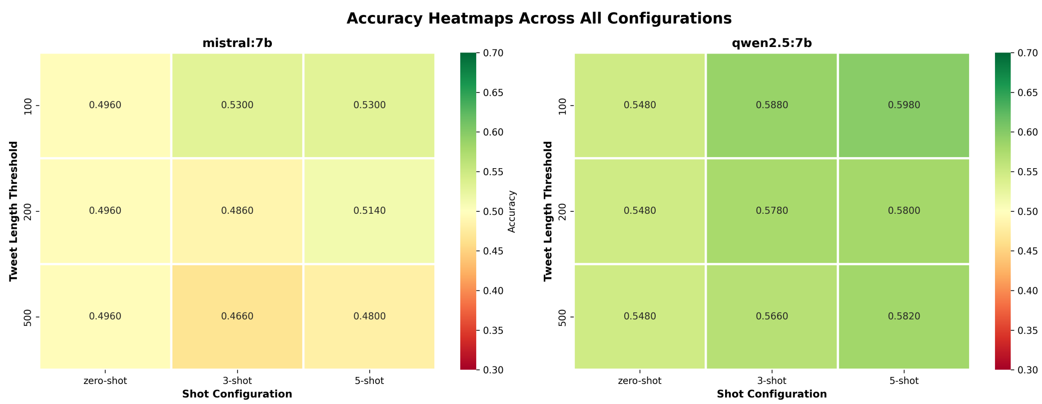

Figure 6: Accuracy heatmap across all configurations for Ollama Models

As a second example, the model was given the following text. **"I had been talking to coach claudia barcomb and coach ali boe for a long time and they both made me feel very welcomed at union"** The model gave the following reasoning behind its labeling decision. **"The phrase "they both made me feel very welcomed" strongly indicates a feeling of **joy**. Being welcomed and feeling positive about a new environment is a core component of experiencing joy. There's no indication of sadness, anger, fear, or surprise.** In this case both the predicted and true labels were joy.

5.3 Metrics

The following metrics were collected for comparison:

- Accuracy
- Balaced accuracy
- F1 weighted
- Matthews correlation coefficient

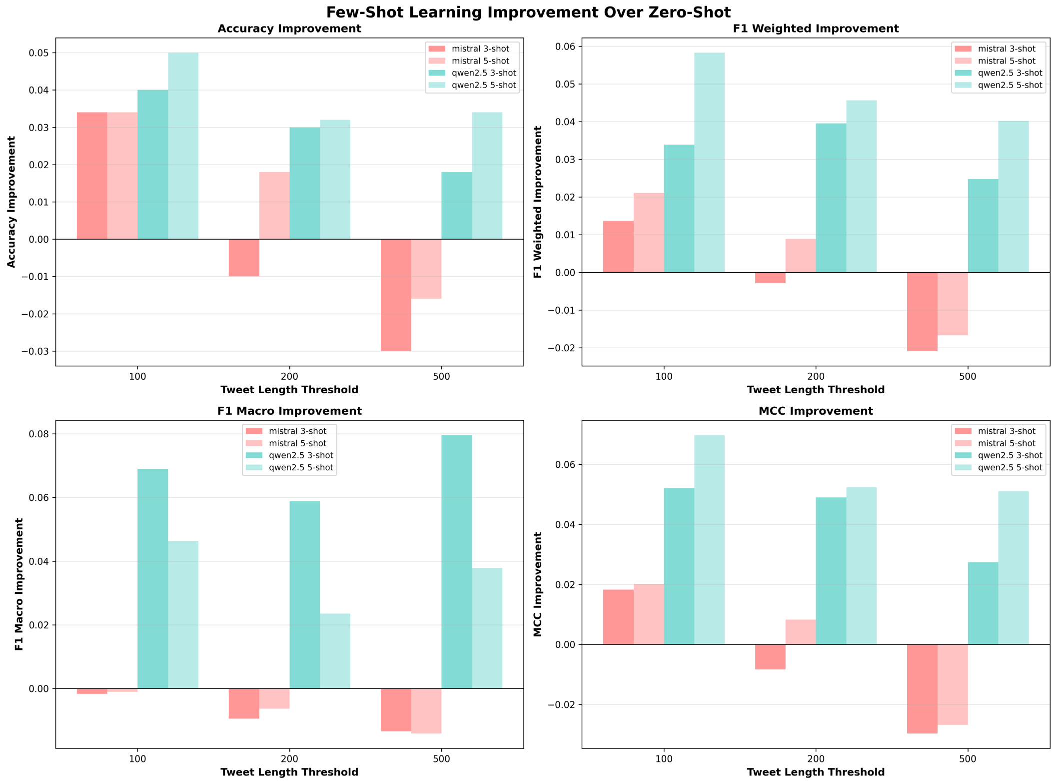

Figure 7: Performance improvements from zero-shot to few-shot on Ollama models

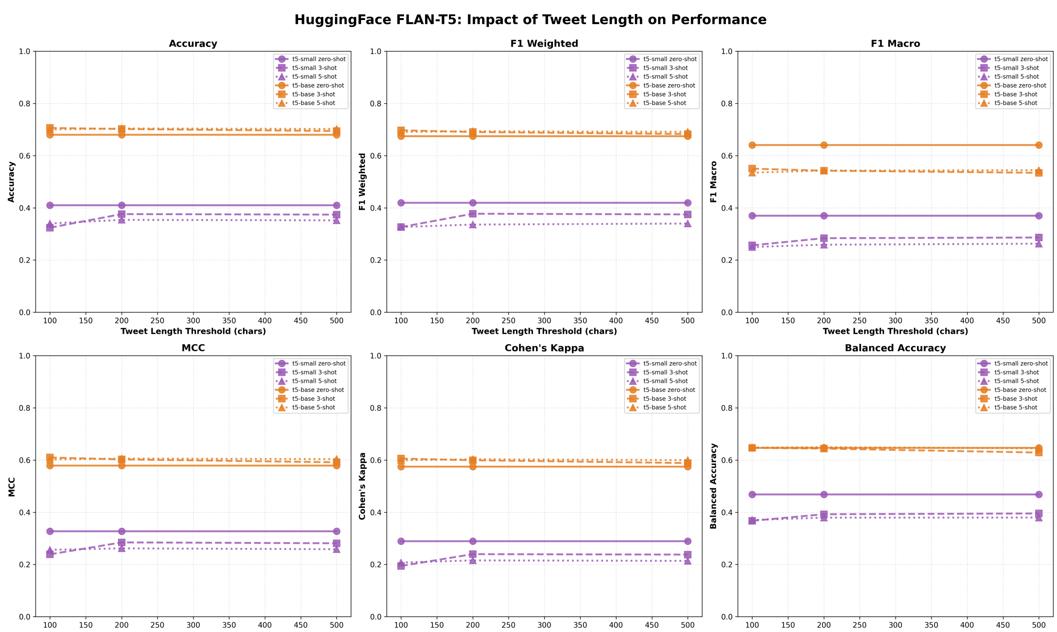

Figure 8: Impact of Tweet Lenght Threshold on Model Performance on HuggingFace Models.

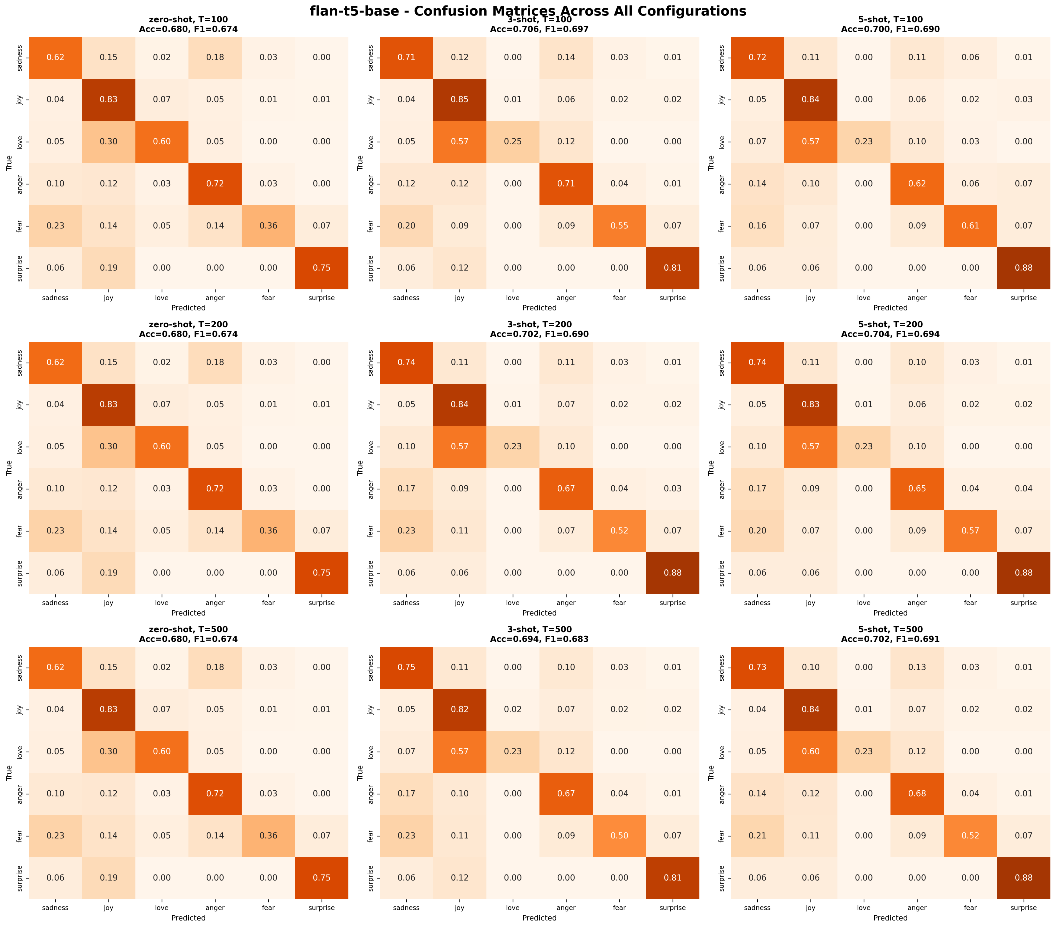

Figure 9: FLAN-T5 Base confusion matrices for all 9 different configurations

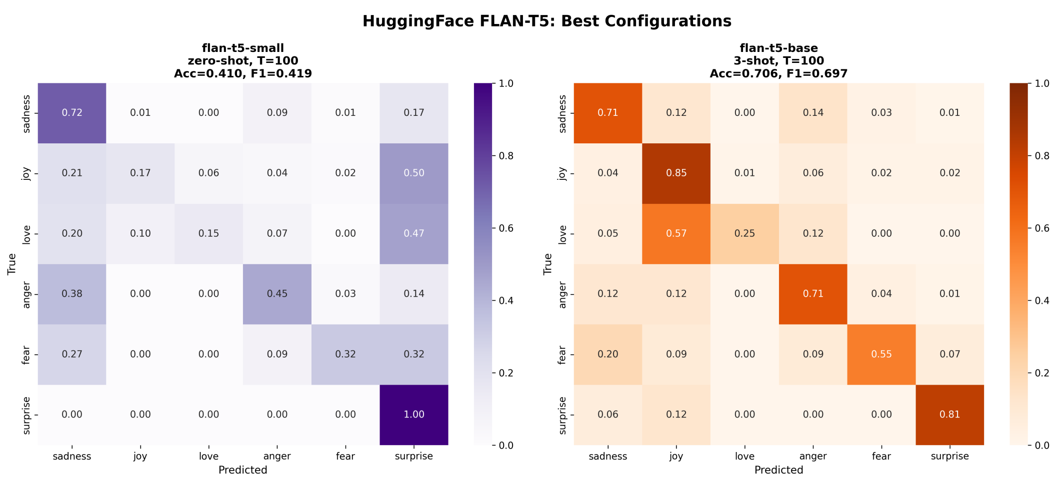

Figure 10: Side-by-side comparison of best configurations for HuggingFace models

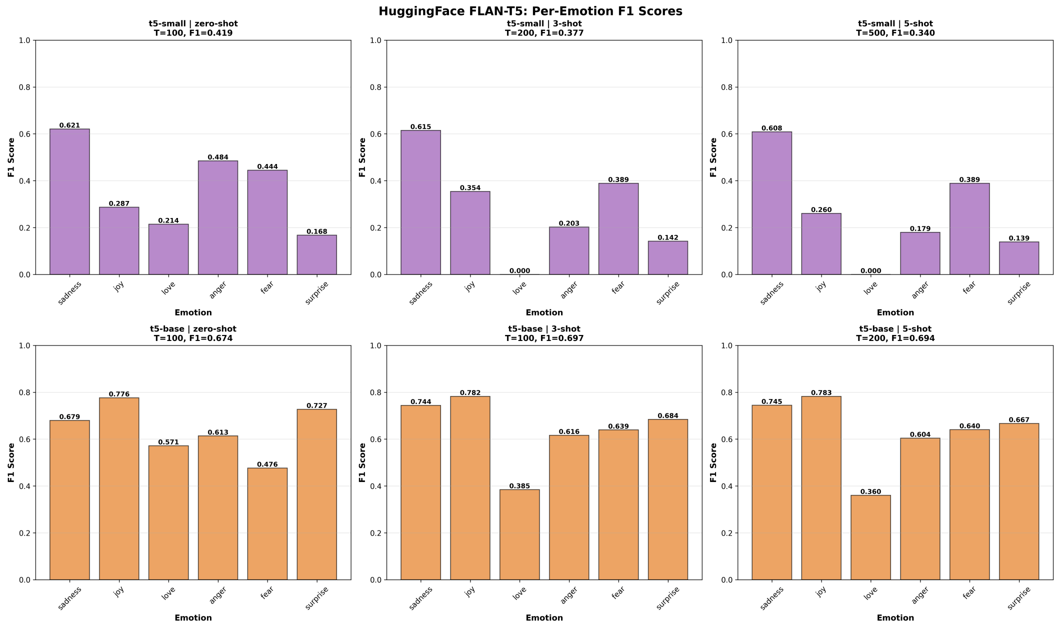

Figure 11: Per-emotion F1-score breakdown across configurations for HuggingFace Models

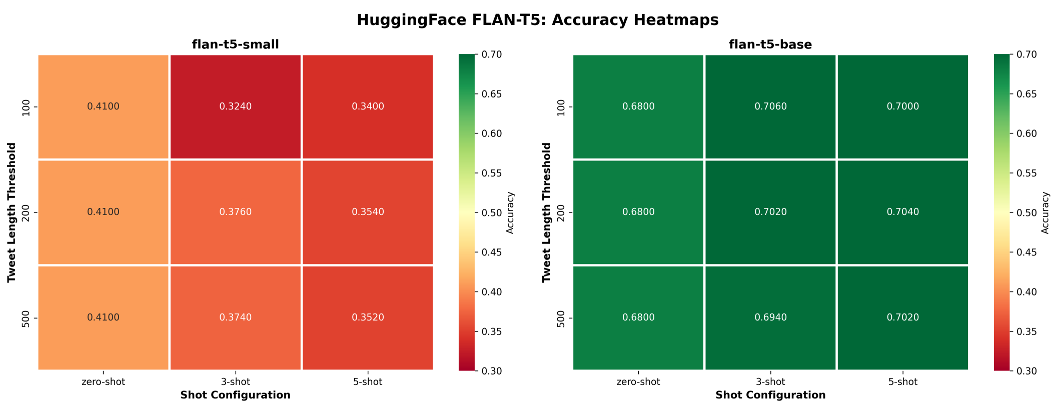

Figure 12: Accuracy heatmap across all configurations for HuggingFace Models

- Cohen kappa score
- Precision
- Recall

5.4 Results

Using the Chain-of-Thought prompt, we found that the Gemma3:4b with CoT achieved the best overall results. Accuracy increased from 36.0% (Gemma3:270m with CoT) to 65.0%, while using the model with more parameters which is aligned with FLAN-T5 model (more parameters on same architecture - better results). Using Chain-of-Thought did not have a major impact on results compared to the model without CoT if we look only at quantitative metrics. Generated reasoning examples still increased trust on models' performance revealing why model predict labels as it did.

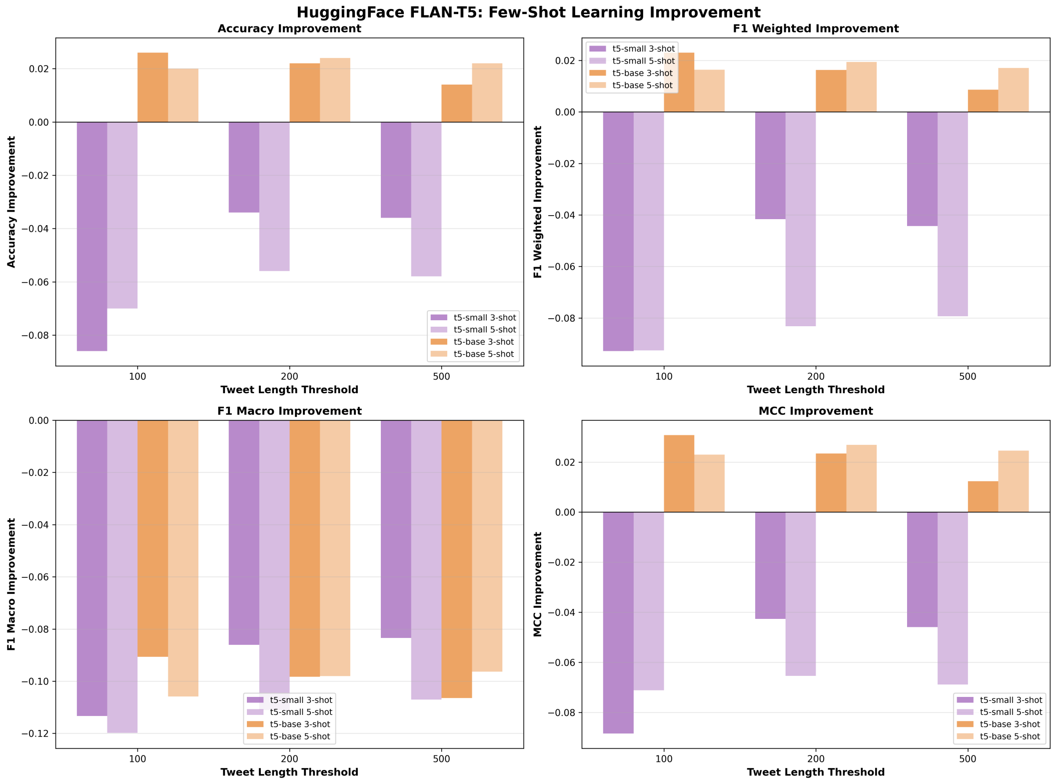

Figure 13: Performance improvements from zero-shot to few-shot on HuggingFace models

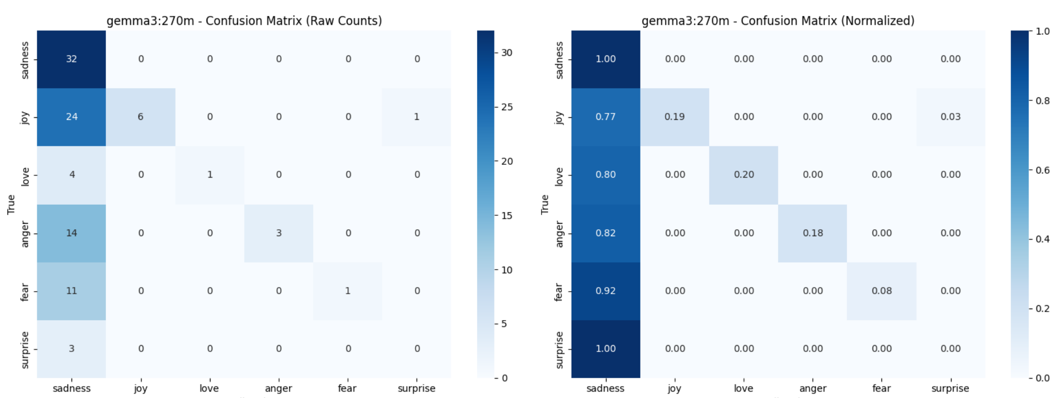

Figure 14: Confusion matrix for Gemma3:270m model.

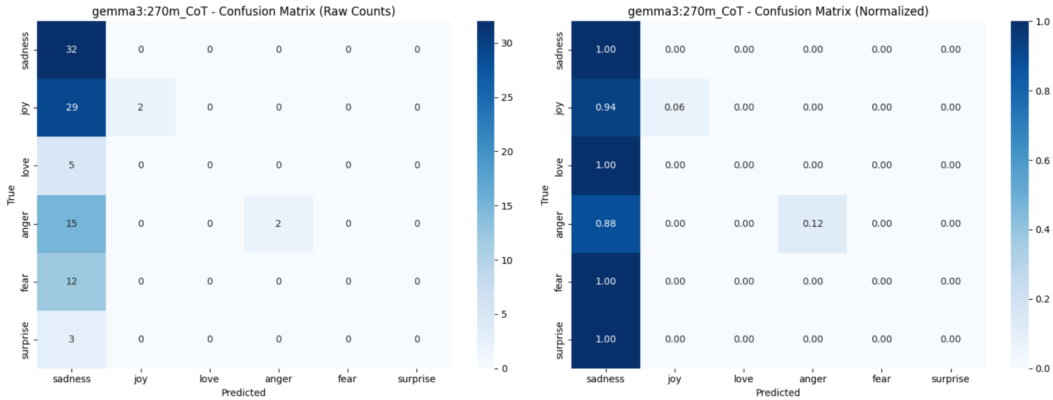

Figure 15: Confusion matrix for Gemma3:270m model using CoT.

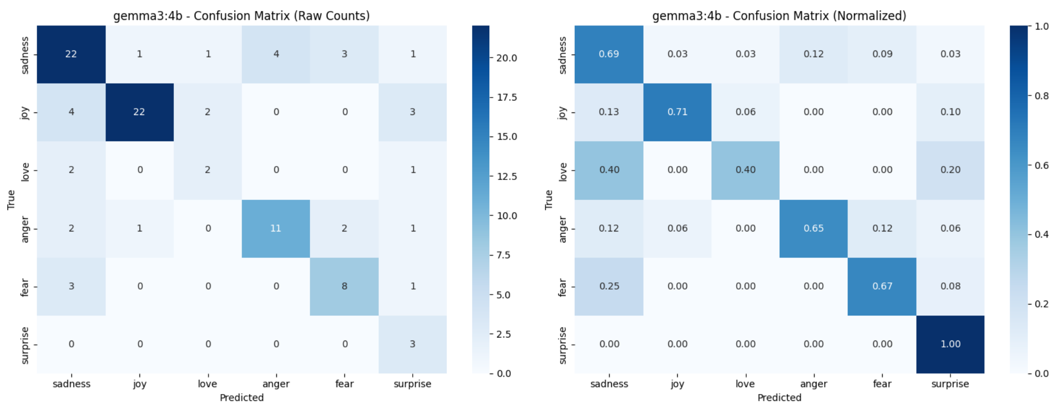

Figure 16: Confusion matrix for Gemma3:4b model.

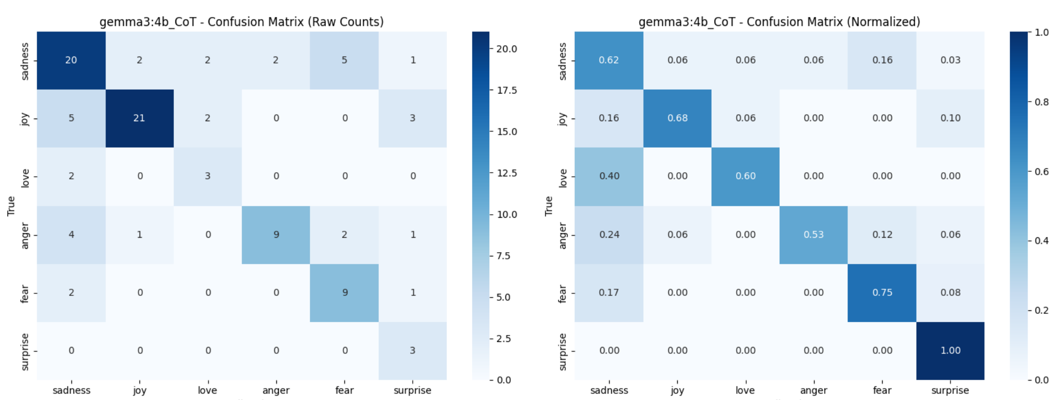

Figure 17: Confusion matrix for Gemma3:4b model using CoT.

Table 3: Model specific metrics with and without CoT.

Model	Accuracy	Balanced accuracy	F1 weighted	Matthews cc.	Cohen kappa
Gemma3:270m	0.43	0.28	0.36	0.30	0.18
Gemma3:270m with CoT	0.36	0.20	0.23	0.18	0.06
Gemma3:4b	0.68	0.69	0.69	0.59	0.59
gemma3:4b with CoT	0.65	0.70	0.66	0.55	0.55

Table 4: Metrics for Gemma3:270m model without the use of CoT.

	Precision	Recall	F1-score
Sadness	0.38	1.00	0.53
Joy	1.00	0.19	0.32
Love	1.00	0.20	0.33
Anger	1.00	0.18	0.30
Fear	1.00	0.08	0.15
Surprise	0.00	0.00	0.00

6 Discussion

This evaluation reveals several insights about the performance of different LLM models in emotion classification tasks, with implications for both model selection and deployment strategies.

The better performance of FLAN-T5 Base over larger Ollama models demonstrates that architectural alignment has a larger impact on the results than the parameter count. The architecture of T5, appears to be better suited for this kind of emotion classification task than autoregressive decoders.

The variation in results with few-shot examples across models indicate a complex relationship between model architecture, capacity, and in-context learning. Seeing how using few-shot with Qwen2.5 7B and FLAN-T5 base had such a positive impact and FLAN-T5 Small the opposite, it seems that good results with few-shot need models that have enough capacity to recognize task patterns from limited examples, overwriting competing patterns from pre-training, and generalizing learned patterns. It seems with smaller models initial examples activate incorrect patterns before sufficient learning therefore giving out bad results. This indicates that smaller models may require alternative strategies such as fine-tuning rather than few-shot prompting.

Results show that 100 character tweets gave out the best outcome across all models. This goes against the thought that longer tweets would give more information, therefore being easier to label

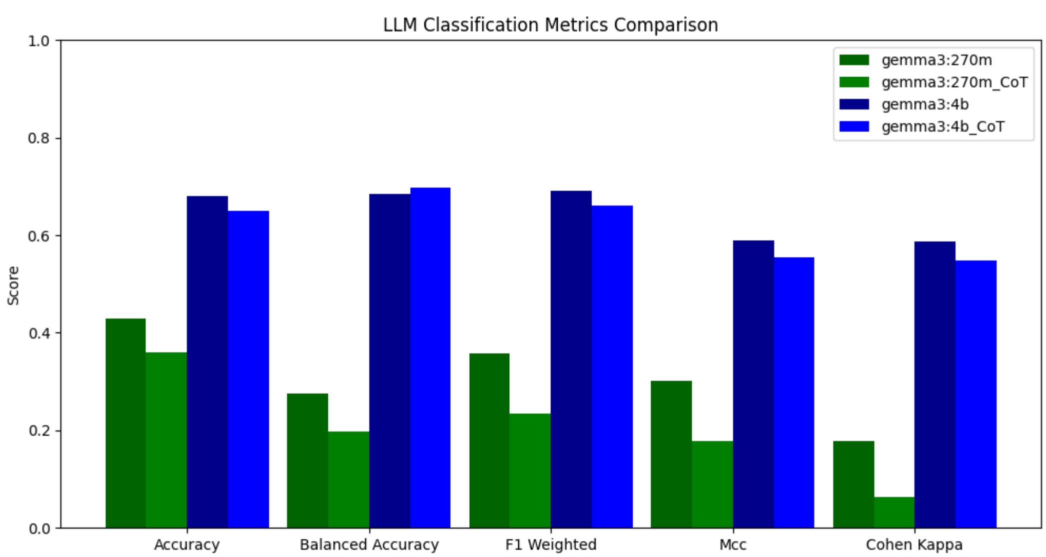

Figure 18: Bar graph comparison of model metrics with and without CoT.

Table 5: Metrics for Gemma3:270m model with CoT.

	Precision	**Recall**	**F1-score**
Sadness	0.33	1.00	0.50
Joy	1.00	0.06	0.12
Love	0.00	0.00	0.00
Anger	1.00	0.12	0.21
Fear	0.00	0.00	0.00
Surprise	0.00	0.00	0.00

Table 6: Metrics for Gemma3:4b model without the use of CoT.

	Precision	**Recall**	**F1-score**
Sadness	0.67	0.69	0.68
Joy	0.92	0.71	0.80
Love	0.40	0.40	0.40
Anger	0.73	0.65	0.69
Fear	0.62	0.67	0.64
Surprise	0.30	1.00	0.46

correctly.There are multiple possible explanations for this. Having shorter tweets can reduce cognitive load, allowing the model to focus on the relevant features of the tweet. Shorter tweets also contain the wanted information in a more dense form, with less "filler-words". Also emotion classification may be mainly dependent on certain markers rather than fuller context. Basically short and simple tweets are more clear to recognize and label, because they are compact with their information and don't have much excess words.

7 Conclusion

While the stratified sample of 500 messages ensures similar class distribution, it may not show the full diversity of emotional expressions in the complete DAIR AI dataset. The relatively low F1-macro scores compared to F1-weighted scores for Ollama models indicate that there are class-wise imbalances in the dataset used. When looking at per-emotion F1-scores, it shows that more represented classes are classified better than those with lower representation.

Chain-of-Thought reasoning employed different pipeline using 100 messages from testing set. Random stratified sampling was not applied, so the resulting class distribution may be different from the few-shot experiment. The results of the classification were still similar.

This shows that effective emotion classification with LLMs depends on the alignment between model architecture, capacity, and learning paradigm. The FLAN-T5 Base model's performance proves to be the best choice among tested models, while the Qwen2.5 7B seemed to be the best Ollama model. The consistent advantage of 100-character few-shot examples gives direction for prompt engineering. The effects of capacity threshold in FLAN-T5 Small indicate that the model should be chosen based on the deployment strategy.

These findings indicate that task-specific model selection and prompt optimization are key to obtaining significant performance benefits opposed to simply choosing the largest model available. This study provides insight to guide on model selection, few-shot prompt design and system architecture decisions.

8 Contribution Distribution

Each group member completed the first steps of running LLMs locally by replicating the steps of a Ollama tutorial linked in the "Final project topics" -file by the teaching assistant. After that areas of responsibility were divided as the following table shows.

Table 7: Metrics for Gemma3:4b model with CoT.

	Precision	**Recall**	**F1-score**
Sadness	0.61	0.62	0.62
Joy	0.88	0.68	0.76
Love	0.43	0.60	0.50
Anger	0.82	0.53	0.64
Fear	0.56	0.75	0.64
Surprise	0.33	1.00	0.50

Table 8: Team Tasks

Team Members	**Tasks**
Salwa Mostafa	Implement the main code for text classification using LLM and presenting the presentation
Severi Raunama	Implement a chain of thought and test explainability to the code
Seyedata Jodeiri Seyedian	Implement few-shot learning to and provide deep insights
Amir Mollazadeh	Writing the technical report and presenting the presentation
Tommi Niemi	Writing the technical report and prepare presentation
Tomi Luukkonen	Writing the technical report and prepare presentation

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